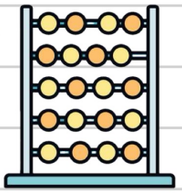
Pseudocode for Code 5 : Max Counters



```
FUNCTION solution (N, A)
    counters ← list of N zeros # makes the row of counters, all starting at 0
    biggest ← 0 # keeps track of the biggest counter so far
    floor ← 0 # lowest any counter is allowed to be
    FOR EACH X IN A # goes through the following steps one at a time
        IF X = N + 1 THEN
            floor ← biggest # raises the floor to the biggest value
        ELSE
            IF counters[X-1] < floor THEN
                counters[X-1] ← floor # if counter is below the floor, it raises it to the floor
            END IF
            counters[X-1] ← counters[X-1] + 1 # adds one to the counter
            IF counters[X-1] > biggest THEN
                biggest ← counters[X-1] # if counter is now biggest, replace biggest
            END IF
        END IF
    END FOR
    FOR C FROM 0 TO N-1 # after all steps, go through each counter once
        IF counters[C] < floor THEN
            counters[C] ← floor # any counter still below floor gets raised to the floor
        END IF
    END FOR
    RETURN counters # gives back final list
END FUNCTION
```